District-Level Crop Yield Prediction for India Using Machine Learning

MSc Data Science Project Presentation

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Dataset: **Apy.csv**

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Research Question and Objectives

Research Question

Can machine learning models accurately predict district-level crop yields across India using historical agricultural production data?

Project Objectives

- Build predictive models using Random Forest and gradient boosting
- Compare ML performance against statistical baselines
- Identify key drivers of yield variation
- Analyze model errors by crop and state

Why This Matters

District-level yield forecasts support agricultural planning and policy decisions. Machine learning can capture complex regional and crop-specific patterns beyond simple historical averages.

Background and Related Work

Key Literature

- Pendergrass et al. (2016): Random Forests for Global and Regional Crop Yield Predictions, PLOS ONE
- IBM Research (2019): Machine learning methodologies for paddy yield estimation in India
- Recent reviews (2023-2025): Tree-based ensembles achieve $R^2 > 0.85$ on tabular agricultural data

Research Gap

Most studies use weather or remote-sensing data. Limited work on pure tabular APY data (Area, Production, Yield only). Need for systematic comparison: statistical baseline vs ML models.

Dataset: India Agricultural Production Yearbook

Data Overview

Source: Government of India open data portal + Kaggle

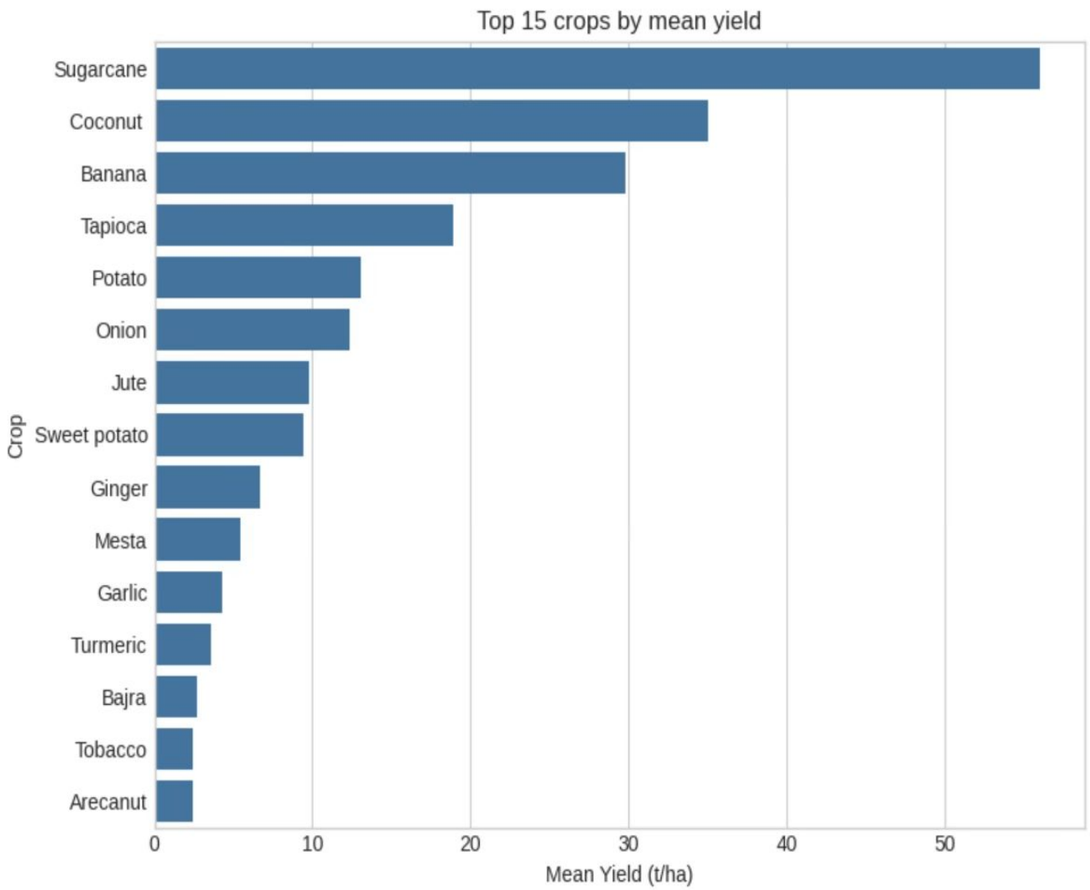
Coverage: 1997–2020, 336,311 district-year-crop records

Features

- Categorical: State (37), District (707), Crop (55), Season (6)
- Numerical: Crop_Year, Area (hectares)
- Target: Yield (tonnes/hectare)

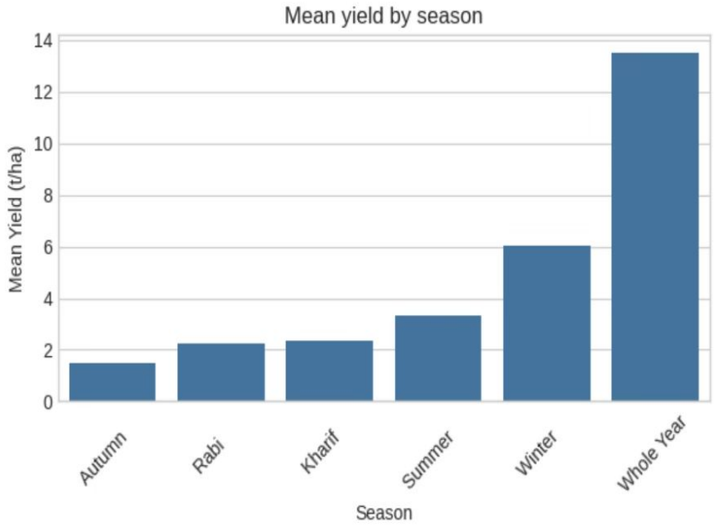
Data Cleaning

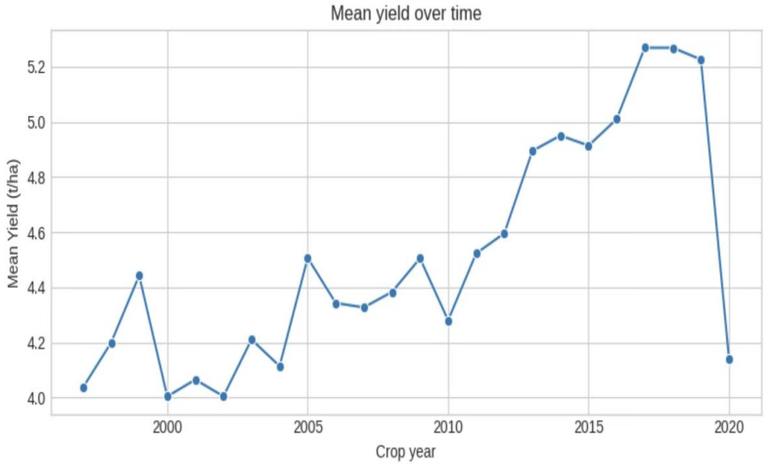
- Removed 9 missing Crop entries
- Filtered non-positive yields (6,091 rows)
- Capped extreme outliers: Yield > 200 t/ha (2,925 rows removed)
- Final dataset: 336,311 observations

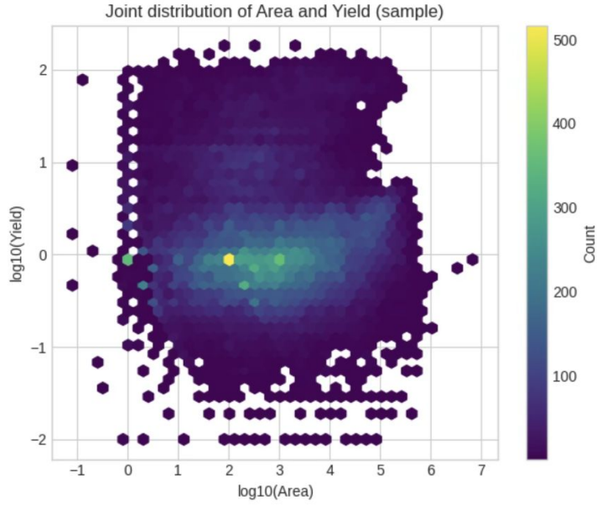


Top-yielding crops: Sugarcane (56 t/ha), Coconut (35 t/ha), and Banana (30 t/ha) far exceed staple cereals like Rice (2.5 t/ha) and Wheat (3.1 t/ha). This 20× variation highlights why crop type is the dominant predictor in our models.

Key Patterns in the Data







Yield by Season

Yield varies by season—Winter and Whole Year have highest yields

Winter and Whole Year seasons achieve 6–14 t/ha, while Kharif/Rabi/Autumn average 1.5–2.3 t/ha. Seasonal differences reflect crop choices (winter crops include high-yield vegetables) and growing conditions.

Yield Over Time

Gradual yield increase from 1997–2019, with a dip in 2020

National average yield rose from 4.0 t/ha (1997) to 5.2 t/ha (2019), reflecting improved seeds, fertilizers, and irrigation. The 2020 dip may indicate COVID-19 disruptions or weather shocks.

Area vs Yield

Yield shows no simple relationship with cultivated area—categorical factors (crop, state) are more important

No clear linear relationship: small and large farms both show wide yield ranges. This confirms that categorical factors (Crop, State, Season) matter more than cultivated area—area has only 5% feature importance.

Model Development Pipeline

01

Temporal Train/Validation/Test Split

Train: 1997–2015 (264,565 records) • Validation: 2016–2017 (34,703 records) • Test: 2018–2020 (37,043 records)

02

Feature Engineering

Ordinal encoding for categorical features (State, District, Crop, Season). Numerical features: Crop_Year, Area. Total features: 6 inputs → Yield (target)

03

Models Trained & Compared

Statistical baseline: Mean-by-(State, Crop) • Linear baseline: Ridge regression (α=1.0) • Random Forest: 200 trees, no max depth, min_samples_leaf=5 • Gradient Boosting: HistGradientBoostingRegressor (depth=10, lr=0.1, 200 iter) • Regularized RF: 300 trees, max_depth=20, min_samples_leaf=10 • Crop-specific RF: Trained only on high-yield crops

04

Hyperparameter Tuning

Compared RF configurations: n_estimators, max_depth, min_samples_leaf. Selected original RF (best test MAE)

Model Performance Comparison

Test Set 2018–2020

Model	Test RMSE (t/ha)	Test MAE (t/ha)	Test R^2
Mean (State, Crop)	4.78	1.67	0.87
Ridge (linear)	12.40	5.87	0.09
Random Forest	4.72	**1.40**	0.87
HistGradientBoosting	**4.52**	1.64	**0.88**
Regularized RF	4.79	1.45	0.86

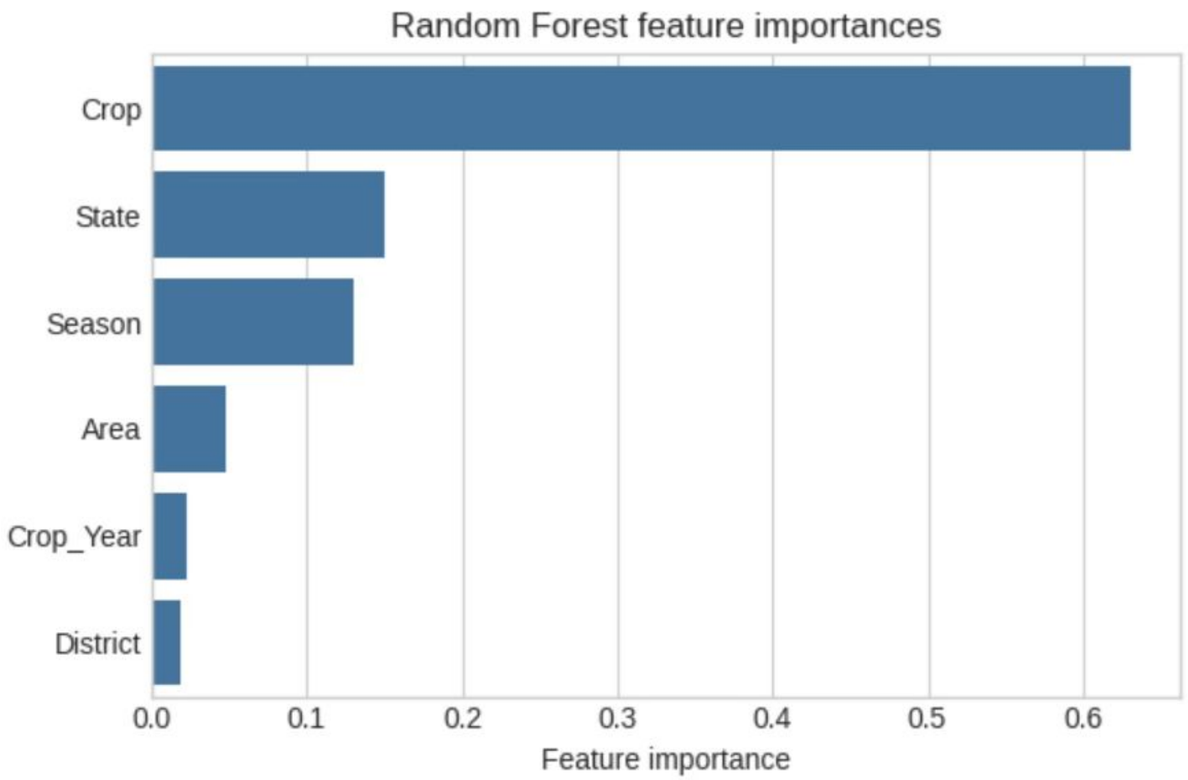
Tree-based models vastly outperform linear regression (R^2 0.87 vs 0.09)

Random Forest achieves lowest MAE (1.40 t/ha), beating statistical baseline

Gradient boosting slightly higher R^2 but worse MAE

Regularization reduced overfitting but didn't improve test performance

What Drives Crop Yield? Feature Importance



Crop dominates (63%), followed by State (15%) and Season (13%). Area and Year contribute <7% combined. This ranking matches agronomic knowledge: what you grow and where matters far more than farm size or temporal trends.

Feature Importance Scores

- Crop (0.63): Dominant factor—sugarcane vs rice vs wheat determines yield
- State (0.15): Regional conditions (climate, soil, infrastructure) matter
- Season (0.13): Growing season affects productivity
- Area (0.05): Farm size has minor impact
- Crop_Year (0.02): Temporal trend is weak predictor
- District (0.02): Within-state variation is small

Interpretation

Model correctly prioritizes agronomically meaningful factors. Aligns with literature: crop type and region drive most yield variation. Validates that the model learns real patterns, not spurious correlations.

Where Does the Model Struggle? Error Analysis

Residual Analysis by Crop

Under-predicted (negative residuals)

- Ginger: –2.25 t/ha
- Oilseeds total: –2.25 t/ha
- Mesta: –1.29 t/ha

Over-predicted (positive residuals)

- Banana: +6.00 t/ha
- Sugarcane: +4.25 t/ha
- Tapioca: +2.96 t/ha
- Potato: +2.03 t/ha

Residual Analysis by State

Under-prediction

- Goa (–2.19)
- Bihar (–0.42)

Over-prediction

- Andhra Pradesh (+3.64)
- Dadra and Nagar Haveli (+5.68)

Why These Errors Occur

- No weather data—drought, floods, temperature extremes not captured
- No soil data—fertility, irrigation, pH not included
- Single national model—high-yield commercial crops have different drivers than cereals
- Aggregation issues—some extreme yields may be reporting errors

Conclusions and Next Steps

Key Findings

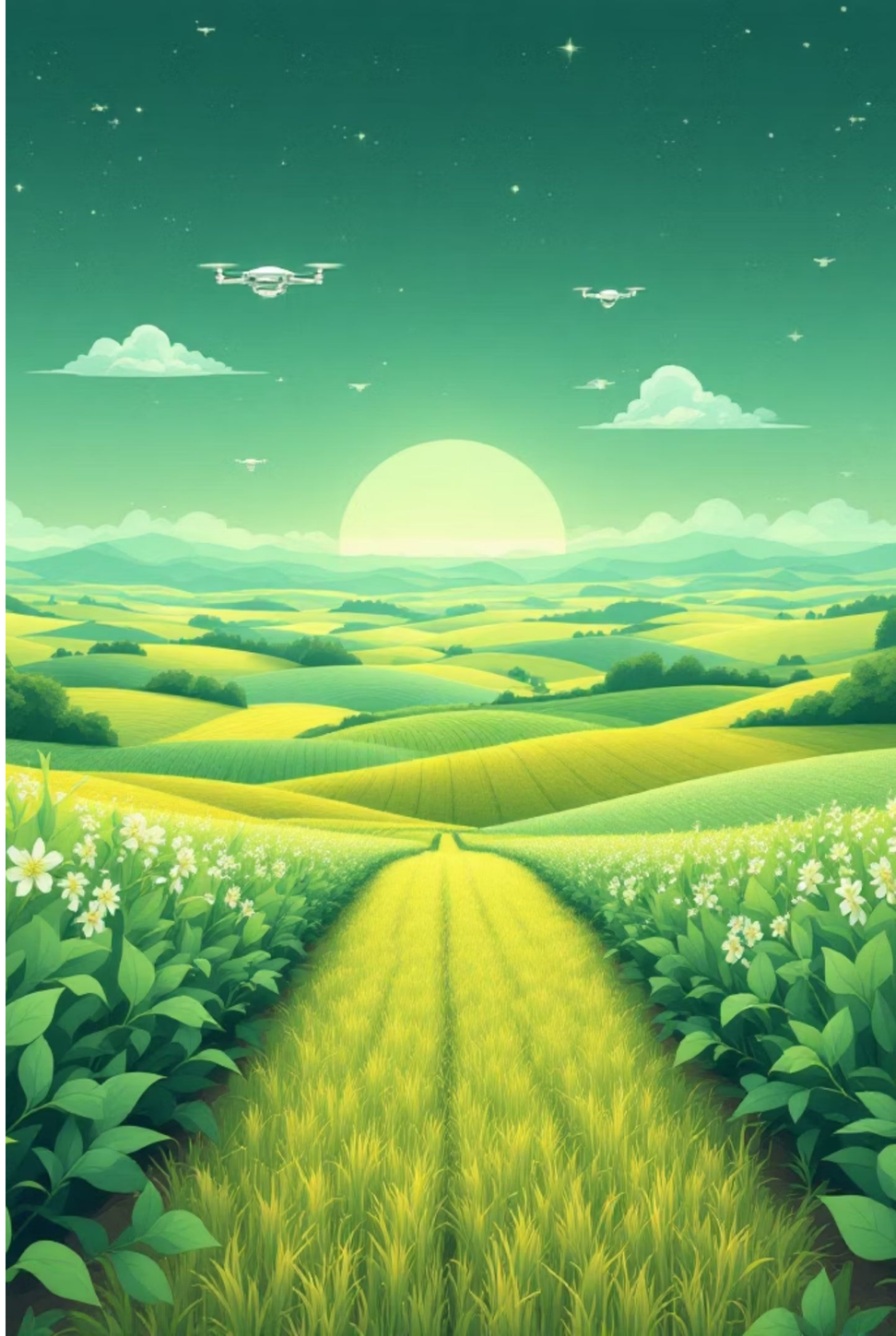
- ML substantially outperforms linear models (R^2 0.87 vs 0.09)
- Random Forest beats baseline (MAE 1.40 vs 1.67 t/ha)
- Crop type and region are dominant predictors
- Systematic biases exist due to missing weather/soil data

Answered Research Question

Yes—machine learning can accurately predict yields (R^2 0.87), but performance is limited without weather and soil data.

Future Work

- Incorporate weather data (rainfall, temperature) from IMD
- Add remote sensing (NDVI, soil moisture)
- Build crop-specific models or hierarchical models
- Deploy as web app for district agricultural offices

Wider Implications

Society: Food security planning, early warning systems. **Technical:** Ensemble methods > linear models. **Commercial:** Agri-tech companies can build yield forecasting SaaS.